HW 1: Free-fall fit

Physics Programming, Week 1 — Core

Submission: upload *one Julia script* and *one figure* (PNG or PDF) on the course Moodle page. This first homework is graded by hand, pass/fail.

Core homework: infer gravity from free-fall measurements

We investigate whether measured fall distances agree with release from rest under approximately uniform gravity. We take distance positive downward, so $y(t) = \frac{1}{2}gt^2 + b$, with $g > 0$ in m/s^2 and offset b in m. Writing $y = at^2 + b$, the fitted slope is a and $g = 2a$; the factor of two comes from the motion equation, not from a software convention.

We work in these stages:

1. Inspect the supplied `free_fall.csv`: identify the time and distance columns, their units, and the number of observations.
2. Predict the shape of y versus t and of y versus t^2 before running the fit.
3. Write a Julia program that reads the file, fits $y = at^2 + b$ with the centered-sum recipe used in the lecture, and computes $g = 2a$, the offset b, the fitted values, and the residuals (measured minus fitted).
4. Make one figure: the measured points as markers with the fitted model overlaid, and every axis labelled with its quantity and units. Save it as **PNG or PDF**.
5. Report the fitted g and the offset with units, say whether the residuals look randomly scattered or patterned, and name one assumption the fit cannot establish.

The program

Write a complete, runnable script (not a function, not a fragment) in a file such as `freefall.jl`. Read the supplied `free_fall.csv` (columns `t_s` in s and `y_m` in m; a CSV is a plain text table with column names in its first row). If the script sits outside the course folder, either point Julia at the course folder and read the file through `PHYSPROG_HOME`, or copy the CSV beside your script. The supplied columns are `t_s` (s) and `y_m` (m). A short script is enough:

```
using CSV, DataFrames, Plots

root = get(ENV, "PHYSPROG_HOME", pwd())    # the course folder
datafile = joinpath(root, "assets", "data", "free_fall.csv")
# Or, if free_fall.csv is beside this script: datafile = "free_fall.csv"
table = CSV.read(datafile, DataFrame)
t_s = table.t_s            # s
y_m = table.y_m            # m

t2 = t_s .^ 2              # s^2
t2_bar = sum(t2) / length(t2)
y_bar  = sum(y_m) / length(y_m)
a = sum((t2 .- t2_bar) .* (y_m .- y_bar)) / sum((t2 .- t2_bar) .^ 2)  #
   m/s^2
b = y_bar - a * t2_bar     # m
g = 2 * a                  # m/s^2
predicted = a .* t2 .+ b   # m
```

```
println("g = ", g, " m/s^2")
println("offset = ", b, " m")

p = plot(t_s, y_m; seriestype = :scatter, label = "measured",
         xlabel = "time (s)", ylabel = "distance (m)")
plot!(p, t_s, predicted; label = "fit")
savefig(p, "freefall_fit.pdf")    # or "freefall_fit.png"
```

Keep this assignment sheet nearby, because it lists the required column names, the figure, and the quantities to report. Rounding a displayed answer must not round the values you use.

Student physics box: downward free fall

We measure downward distance y from the release point, so downward is positive. For a dropped object over a short distance with negligible air drag and approximately constant gravitational acceleration, $y = at^2 + b$, where t is seconds, y metres, a is m/s^2, and $g = 2a$ is the positive gravity magnitude. At $t = 0$, $y = b$; an ideal origin has $b = 0$. A timing or distance offset, drag, or changing acceleration can produce residual structure or a nonzero intercept.

Required evidence

Run your program on the supplied CSV and submit the script and the figure. Report g, the offset b, the units, whether the residuals look randomly scattered or patterned, and the name of the figure file. Do not claim precision beyond the measurements.

How this homework is checked

This first homework is graded by hand on Moodle, pass/fail. The teaching staff run your script and look at the figure: the script reads the supplied file and fits $y = at^2 + b$, and the figure shows the measured points, the fitted line, and labelled axes with units. Correct output and a clear figure pass. From Week 2 the homework returns to automatic testing.